

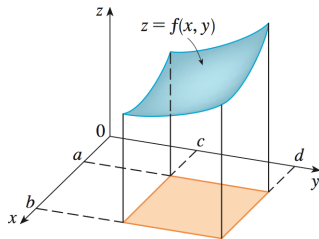
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function

Ex: $f(x) = x^2$ is a scalar function since
 $x \in \mathbb{R}$ and $f(x) \in \mathbb{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since
 $(x, y) \in \mathbb{R}^2$ and $f(x, y) \in \mathbb{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in \mathbb{R}^3$ and $f(x, y, z) \in \mathbb{R}$

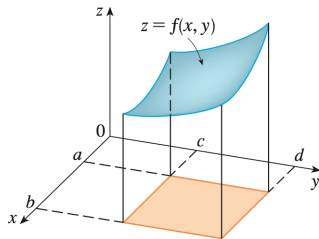
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since
 $x \in \mathbb{R}$ and $f(x) \in \mathbb{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since
 $(x, y) \in \mathbb{R}^2$ and $f(x, y) \in \mathbb{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in \mathbb{R}^3$ and $f(x, y, z) \in \mathbb{R}$

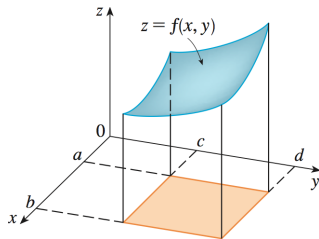
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbb{R}$ and $f(x) \in \mathbb{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since
 $(x, y) \in \mathbb{R}^2$ and $f(x, y) \in \mathbb{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in \mathbb{R}^3$ and $f(x, y, z) \in \mathbb{R}$

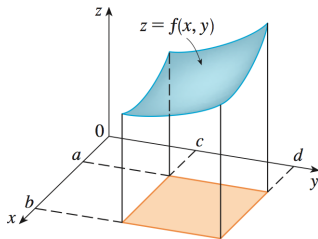
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since
 $(x, y) \in$ and $f(x, y) \in$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in$ and $f(x, y, z) \in$

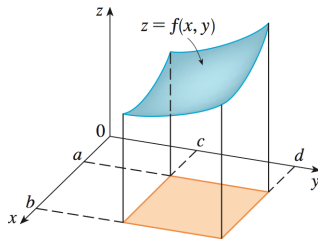
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since
 $(x, y) \in$ and $f(x, y) \in$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in$ and $f(x, y, z) \in$

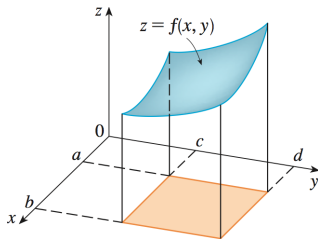
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since
 $(x, y) \in$ and $f(x, y) \in$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in$ and $f(x, y, z) \in$

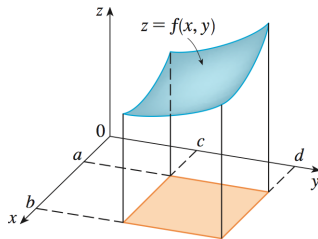
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since $f(1, 4) = (1) - (1)\sqrt{(4)} = -1 \leftarrow (\text{scalar})$
 $(x, y) \in$ and $f(x, y) \in$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in$ and $f(x, y, z) \in$

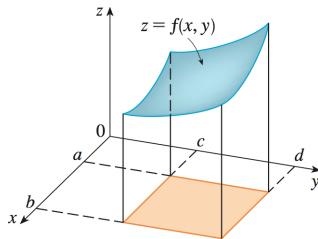
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since $f(1, 4) = (1) - (1)\sqrt{(4)} = -1 \leftarrow (\text{scalar})$
 $(x, y) \in \mathbf{R}^2$ and $f(x, y) \in \mathbf{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in \mathbf{R}^3$ and $f(x, y, z) \in \mathbf{R}$

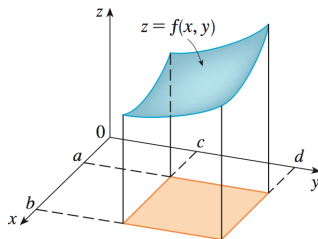
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since $f(1, 4) = (1) - (1)\sqrt{(4)} = -1 \leftarrow (\text{scalar})$
 $(x, y) \in \mathbf{R}^2$ and $f(x, y) \in \mathbf{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in \mathbf{R}^3$ and $f(x, y, z) \in \mathbf{R}$

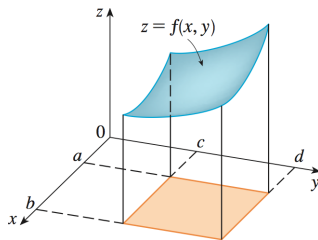
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since $f(1, 4) = (1) - (1)\sqrt{(4)} = -1 \leftarrow (\text{scalar})$
 $(x, y) \in \mathbf{R}^2$ and $f(x, y) \in \mathbf{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since $f(-1, 0, \pi) = \frac{\cos(\pi)}{(-1)} - (-1) = 2 \leftarrow (\text{scalar})$
 $(x, y, z) \in$ and $f(x, y, z) \in$

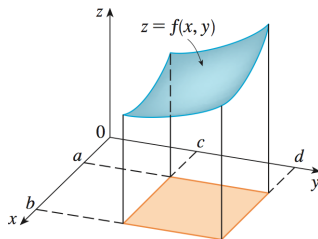
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since $f(1, 4) = (1) - (1)\sqrt{(4)} = -1 \leftarrow (\text{scalar})$
 $(x, y) \in \mathbf{R}^2$ and $f(x, y) \in \mathbf{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since $f(-1, 0, \pi) = \frac{\cos(\pi)}{(-1)} - (-1) = 2 \leftarrow (\text{scalar})$
 $(x, y, z) \in \mathbf{R}^3$ and $f(x, y, z) \in \mathbf{R}$

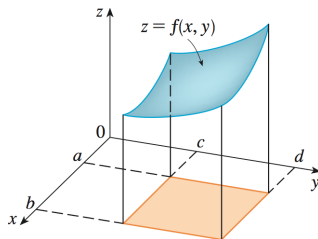
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \leftarrow (\text{scalar})$
 $x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since $f(1, 4) = (1) - (1)\sqrt{(4)} = -1 \leftarrow (\text{scalar})$
 $(x, y) \in \mathbf{R}^2$ and $f(x, y) \in \mathbf{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since $f(-1, 0, \pi) = \frac{\cos(\pi)}{(-1)} - (-1) = 2 \leftarrow (\text{scalar})$
 $(x, y, z) \in \mathbf{R}^3$ and $f(x, y, z) \in \mathbf{R}$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f =$$

$$\nabla g =$$

Ex: $f(x, y) = xy + y$. Find ∇f

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] =$$

$$\nabla g =$$

Ex: $f(x, y) = xy + y$. Find ∇f

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g =$$

Ex: $f(x, y) = xy + y$. Find ∇f

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] =$$

Ex: $f(x, y) = xy + y$. Find ∇f

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} =$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and}$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} =$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2 + y)$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2 + y)$$

$$\implies \nabla f(P) = \left[\frac{\partial f}{\partial x}(P), \frac{\partial f}{\partial y}(P), \frac{\partial f}{\partial z}(P) \right] =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2 + y)$$

$$\implies \nabla f(P) = \left[\frac{\partial f}{\partial x}(P), \frac{\partial f}{\partial y}(P), \frac{\partial f}{\partial z}(P) \right] = \left[\quad, \quad, \quad \right] =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2 + y)$$

$$\implies \nabla f(P) = \left[\frac{\partial f}{\partial x}(P), \frac{\partial f}{\partial y}(P), \frac{\partial f}{\partial z}(P) \right] = \left[\frac{(2)(1)^2}{(0)(1)^2 + (1)}, \quad , \quad \right] =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2 + y)$$

$$\implies \nabla f(P) = \left[\frac{\partial f}{\partial x}(P), \frac{\partial f}{\partial y}(P), \frac{\partial f}{\partial z}(P) \right] = \left[\frac{(2)(1)^2}{(0)(1)^2 + (1)}, \frac{(2)(2(0)(1) + 1)}{(0)(1)^2 + (1)}, \right] =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2 + y)$$

$$\implies \nabla f(P) = \left[\frac{\partial f}{\partial x}(P), \frac{\partial f}{\partial y}(P), \frac{\partial f}{\partial z}(P) \right] = \left[\frac{(2)(1)^2}{(0)(1)^2 + (1)}, \frac{(2)(2(0)(1) + 1)}{(0)(1)^2 + (1)}, \ln((0)(1)^2 + (1)) \right] =$$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right] = [f_x, f_y] = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] = [g_x, g_y, g_z] = g_x \mathbf{i} + g_y \mathbf{j} + g_z \mathbf{k}$$

Ex: $f(x, y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \implies \quad \nabla f = [y, x + 1]$$

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

$$\frac{\partial f}{\partial x} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial x}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot y^2 = \frac{zy^2}{xy^2 + y}$$

$$\frac{\partial f}{\partial y} = z \frac{1}{xy^2 + y} \cdot \frac{\partial}{\partial y}(xy^2 + y) = z \frac{1}{xy^2 + y} \cdot (2xy + 1) = \frac{z(2xy + 1)}{xy^2 + y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2 + y)$$

$$\implies \nabla f(P) = \left[\frac{\partial f}{\partial x}(P), \frac{\partial f}{\partial y}(P), \frac{\partial f}{\partial z}(P) \right] = \left[\frac{(2)(1)^2}{(0)(1)^2 + (1)}, \frac{(2)(2(0)(1) + 1)}{(0)(1)^2 + (1)}, \ln((0)(1)^2 + (1)) \right] = [2, 2, 0]$$

Prove the following two properties of gradients:

$$\text{Recall: } \nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$$

1. $\nabla(f + g) = \nabla f + \nabla g$

Proof : $\nabla(f + g) =$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

Proof : $\nabla(f + g) = \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient})$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

Proof : $\nabla(f + g) = \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient})$

$$= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right]$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\text{Proof : } \nabla(fg) = \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient})$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient}) \\ &= \left[f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z \right] \end{aligned}$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient}) \\ &= \left[f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z \right] \quad (\text{Product Rule} \times 3) \end{aligned}$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient}) \\ &= \left[f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z \right] \quad (\text{Product Rule} \times 3) \\ &= \left[f_x \cdot g, f_y \cdot g, f_z \cdot g \right] + \left[f \cdot g_x, f \cdot g_y, f \cdot g_z \right] \end{aligned}$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient}) \\ &= \left[f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z \right] \quad (\text{Product Rule} \times 3) \\ &= \left[f_x \cdot g, f_y \cdot g, f_z \cdot g \right] + \left[f \cdot g_x, f \cdot g_y, f \cdot g_z \right] \quad (\text{Vector Addition}) \end{aligned}$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient}) \\ &= \left[f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z \right] \quad (\text{Product Rule} \times 3) \\ &= \left[f_x \cdot g, f_y \cdot g, f_z \cdot g \right] + \left[f \cdot g_x, f \cdot g_y, f \cdot g_z \right] \quad (\text{Vector Addition}) \\ &= \left[f_x, f_y, f_z \right] \cdot g + f \cdot \left[g_x, g_y, g_z \right] \end{aligned}$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient}) \\ &= \left[f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z \right] \quad (\text{Product Rule} \times 3) \\ &= \left[f_x \cdot g, f_y \cdot g, f_z \cdot g \right] + \left[f \cdot g_x, f \cdot g_y, f \cdot g_z \right] \quad (\text{Vector Addition}) \\ &= \left[f_x, f_y, f_z \right] \cdot g + f \cdot \left[g_x, g_y, g_z \right] \quad (\text{Scalar} \cdot \text{Vector}) \end{aligned}$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient}) \\ &= \left[f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z \right] \quad (\text{Product Rule} \times 3) \\ &= \left[f_x \cdot g, f_y \cdot g, f_z \cdot g \right] + \left[f \cdot g_x, f \cdot g_y, f \cdot g_z \right] \quad (\text{Vector Addition}) \\ &= \left[f_x, f_y, f_z \right] \cdot g + f \cdot \left[g_x, g_y, g_z \right] \quad (\text{Scalar} \cdot \text{Vector}) \\ &= \nabla f \cdot g + f \cdot \nabla g \end{aligned}$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$

1. $\nabla(f + g) = \nabla f + \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(f + g) &= \left[\frac{\partial}{\partial x}(f + g), \frac{\partial}{\partial y}(f + g), \frac{\partial}{\partial z}(f + g) \right] \quad (\text{Defn of Gradient}) \\ &= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z} \right] \quad (\text{Addition Rule} \times 3) \\ &= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right] \quad (\text{Vector Addition}) \\ &= \nabla f + \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$\begin{aligned} \text{Proof : } \nabla(fg) &= \left[(f \cdot g)_x, (f \cdot g)_y, (f \cdot g)_z \right] \quad (\text{Defn of Gradient}) \\ &= \left[f_x \cdot g + f \cdot g_x, f_y \cdot g + f \cdot g_y, f_z \cdot g + f \cdot g_z \right] \quad (\text{Product Rule} \times 3) \\ &= \left[f_x \cdot g, f_y \cdot g, f_z \cdot g \right] + \left[f \cdot g_x, f \cdot g_y, f \cdot g_z \right] \quad (\text{Vector Addition}) \\ &= \left[f_x, f_y, f_z \right] \cdot g + f \cdot \left[g_x, g_y, g_z \right] \quad (\text{Scalar} \cdot \text{Vector}) \\ &= \nabla f \cdot g + f \cdot \nabla g \quad (\text{Defn of Gradient} \times 2) \end{aligned}$$

III. Additional Concepts

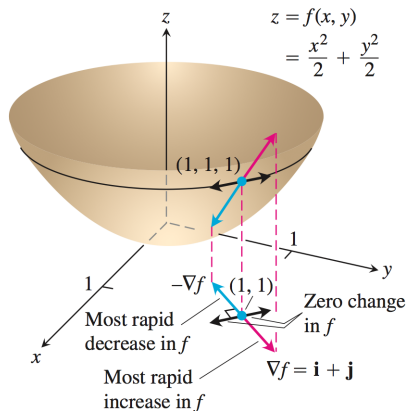
Thm: $\nabla f(P)$ points in the direction of

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \quad \text{so we also have} \quad D_{\mathbf{u}}f =$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$



III. Additional Concepts

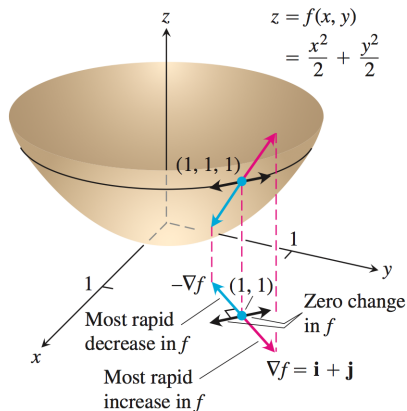
Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \quad \text{so we also have} \quad D_{\mathbf{u}}f =$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$



III. Additional Concepts

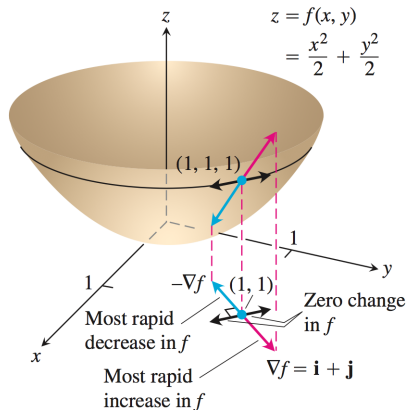
Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \quad \text{so we also have} \quad D_{\mathbf{u}}f =$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$



III. Additional Concepts

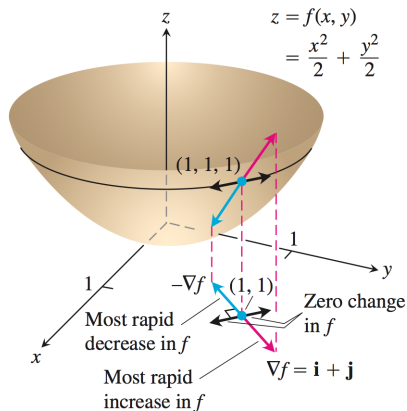
Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \quad \text{so we also have} \quad D_{\mathbf{u}}f =$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$



III. Additional Concepts

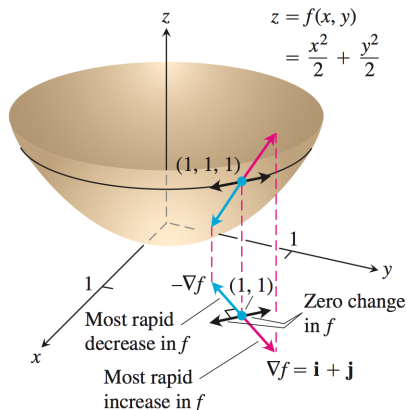
Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \quad \text{so we also have} \quad D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f =$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$



III. Additional Concepts

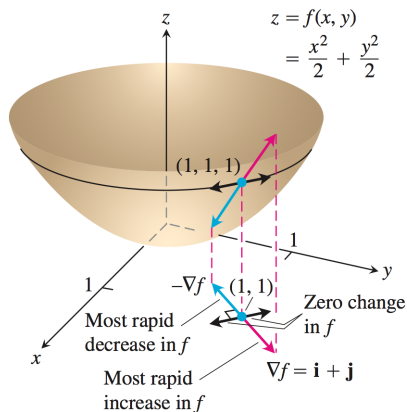
Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \quad \text{so we also have} \quad D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f = |\mathbf{u}| |\nabla f| \cos \theta$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$



III. Additional Concepts

Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

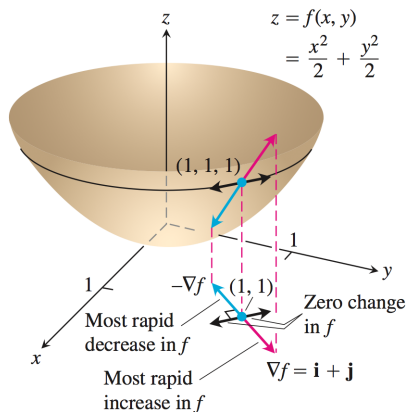
Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \quad \text{so we also have} \quad D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f = |\mathbf{u}| |\nabla f| \cos \theta$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$

$$\nabla f = [x, y]$$



III. Additional Concepts

Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

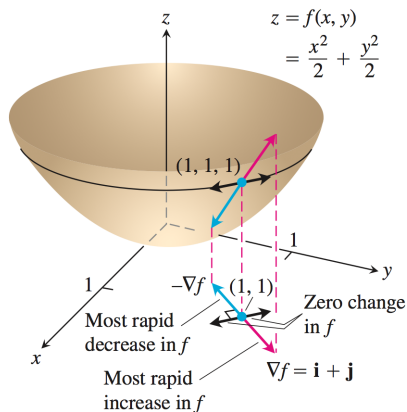
Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \quad \text{so we also have} \quad D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f = |\mathbf{u}| |\nabla f| \cos \theta$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$

$\nabla f = [x, y]$ and since $|\mathbf{u}| = 1$



III. Additional Concepts

Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

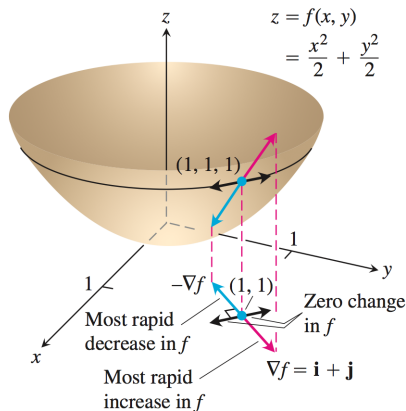
$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \quad \text{so we also have} \quad D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f = |\mathbf{u}| |\nabla f| \cos \theta$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$

$\nabla f = [x, y]$ and since $|\mathbf{u}| = 1$

$$\Rightarrow D_{\mathbf{u}}f = [x, y] \cdot \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right] = \frac{1}{2}x + \frac{\sqrt{3}}{2}y$$



III. Additional Concepts

Thm: $\nabla f(P)$ points in the direction of **maximum increase of f** at the point P .

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \quad \text{so we also have} \quad D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f = |\mathbf{u}| |\nabla f| \cos \theta$$

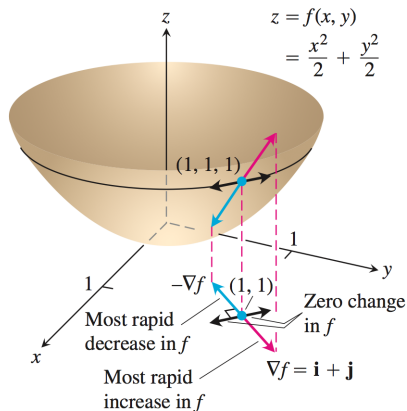
Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$

$\nabla f = [x, y]$ and since $|\mathbf{u}| = 1$

$$\Rightarrow D_{\mathbf{u}}f = [x, y] \cdot \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right] = \frac{1}{2}x + \frac{\sqrt{3}}{2}y$$

$$\Rightarrow D_{\mathbf{u}}f(P) = \frac{1}{2}(1) + \frac{\sqrt{3}}{2}(1) = \frac{1 + \sqrt{3}}{2}$$



Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER:

REASON:

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T =$$

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y]$$

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)]$$

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

$$\implies |\nabla T(P)| = \|[-4, 13]\| = \sqrt{(-4)^2 + (13)^2} = \sqrt{185} \approx 13.6$$

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

$$\implies |\nabla T(P)| = \|[-4, 13]\| = \sqrt{(-4)^2 + (13)^2} = \sqrt{185} \approx 13.6$$

$\implies$ Max rate of change at P is:

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

$$\implies |\nabla T(P)| = |[-4, 13]| = \sqrt{(-4)^2 + (13)^2} = \sqrt{185} \approx 13.6$$

$$\implies \text{Max rate of change at } P \text{ is: } \sqrt{185}$$

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/ft$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

$$\implies |\nabla T(P)| = \|[-4, 13]\| = \sqrt{(-4)^2 + (13)^2} = \sqrt{185} \approx 13.6$$

$$\implies \text{Max rate of change at } P \text{ is: } \sqrt{185}$$

$$\implies \text{Min rate of change at } P \text{ is:}$$

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/ft$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

$$\implies |\nabla T(P)| = \|[-4, 13]\| = \sqrt{(-4)^2 + (13)^2} = \sqrt{185} \approx 13.6$$

$$\implies \text{Max rate of change at } P \text{ is: } \sqrt{185}$$

$$\implies \text{Min rate of change at } P \text{ is: } -\sqrt{185}$$

Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

$$\implies |\nabla T(P)| = \|[-4, 13]\| = \sqrt{(-4)^2 + (13)^2} = \sqrt{185} \approx 13.6$$

$$\implies \text{Max rate of change at } P \text{ is: } \sqrt{185}$$

$$\implies \text{Min rate of change at } P \text{ is: } -\sqrt{185}$$

Therefore, since 14 does not lie within $[\sqrt{185}, -\sqrt{185}]$, the rate of change of the temperature can never be equal to $14^\circ\text{C}/\text{ft}$